Software Engineering

LAB 3: UML CLASS DIAGRAMS

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Learning Objectives

- This guide will show you how to understand, plan, and create your own class diagrams.
- Task to do

- Class diagrams are one of the most useful types of diagrams in UML as they clearly map out the structure of a particular system by modelling its classes, attributes, operations, and relationships between objects.

WHAT IS A CLASS DIAGRAM IN UML

- The Unified Modelling Language (UML) can help you model systems in various ways. One of the most popular UML diagram types is the class diagram. Popular among software engineers to document software architecture, class diagrams are a type of structure diagram because they describe what must be present in the system being modelled.

Benefits of Class Diagrams

- Illustrate data models for information systems,
- Better understand the general overview of the schematics of an application
- Visually express any specific needs of a system
- Create detailed charts that highlight any specific code needed to be programmed and implemented
- Provide an implementation-independent description of types used in a system

The Standard Class Diagram Is Composed Of Three Sections

- **Upper section**: Contains the name of the class.
- **Middle section:** Contains the attributes of the class.
- **Bottom section:** Includes class operations (methods).

Member Access Modifiers

- Public (+)
- Private (-)
- Protected (#)
- Package (~)
- Derived (/)
- Static (underlined)

Real-World" example involving a **Library System**

```java
// Parent Class
public class LibraryItem {
    protected String title;
    protected String itemID;

    public void checkout() {
        // Code to checkout item
    }
}

// Subclass (Inheritance)
public class Book extends LibraryItem {
    private String author;
    public int pages;

    public String getAuthor() {
        return author;
    }
}

// Container Class (Aggregation)
public class Library {
    public String name;
    public List<Book> books; // Library has a collection of Books

    public void addBook(Book newBook) {
        books.add(newBook);
    }
}
```

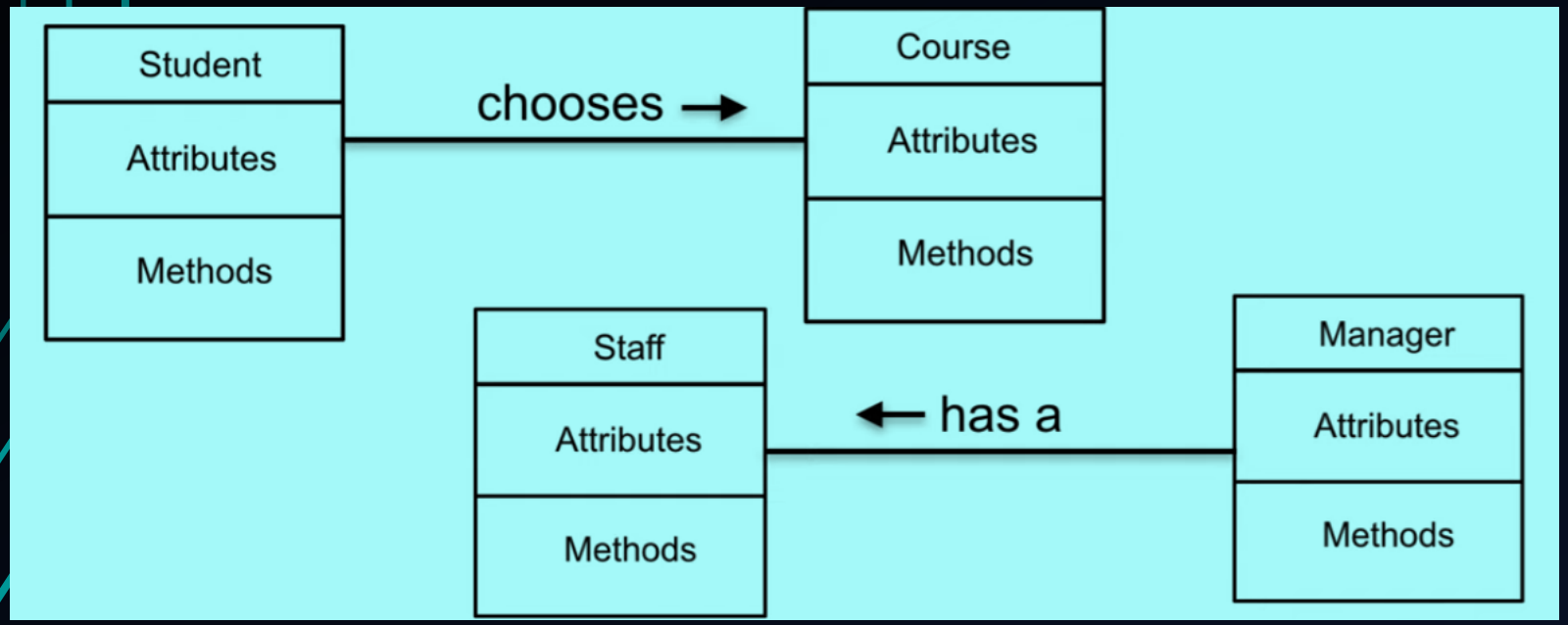
Real-World" example involving a **Library System**

Class	Attributes	Methods
LibraryItem	# title : String # itemID : String	+ checkout() : void
Book	- author : String + pages : int	+ getAuthor() : String
Library	+ name : String	+ addBook(newBook : Book) : void

Visibility

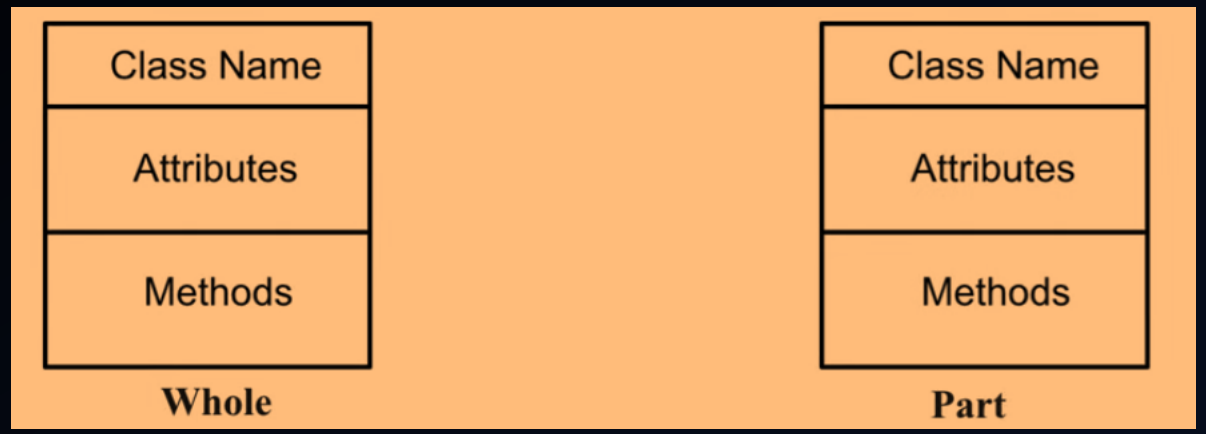
Access Right	public (+)	private (-)	protected (#)	Package (~)
Members of the same class	yes	yes	yes	yes
Members of derived classes	yes	no	yes	yes
Members of any other class	yes	no	no	in same

Association



Description	Multiplicity
Exactly One	1
Zero or More	0..*
One or More	1..*
Specified Range	2..4
Multiple Disjoint	2,4..6,8
One or Other	2,4

Aggregation



For Example

- College & Student
- Library & Book
- Room & Walls
- Employee list & Employee
- Car & Engine

Aggregation: The "Part" can exist without the "Whole."

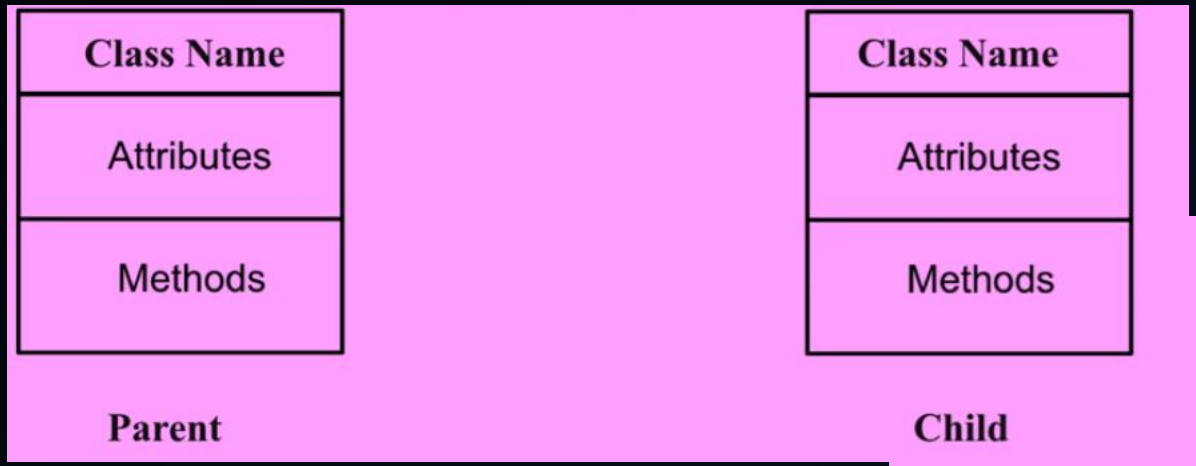
Composition

For Example

- Building and Rooms
- Browser & Tab
- Person & Head

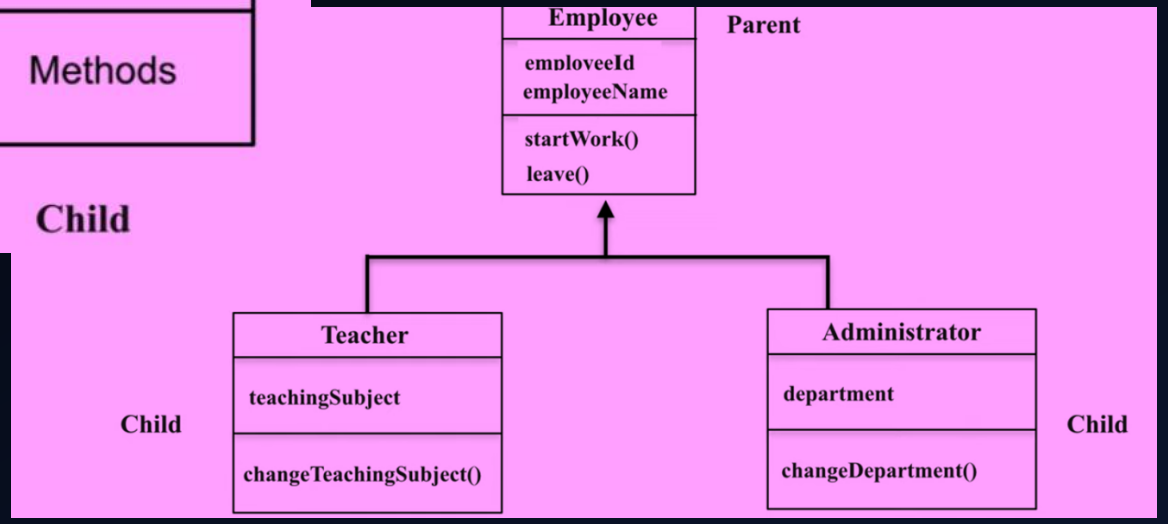
Aggregation: The "Part" can exist without the "Whole."

Inheritance (Generalization)



Example:

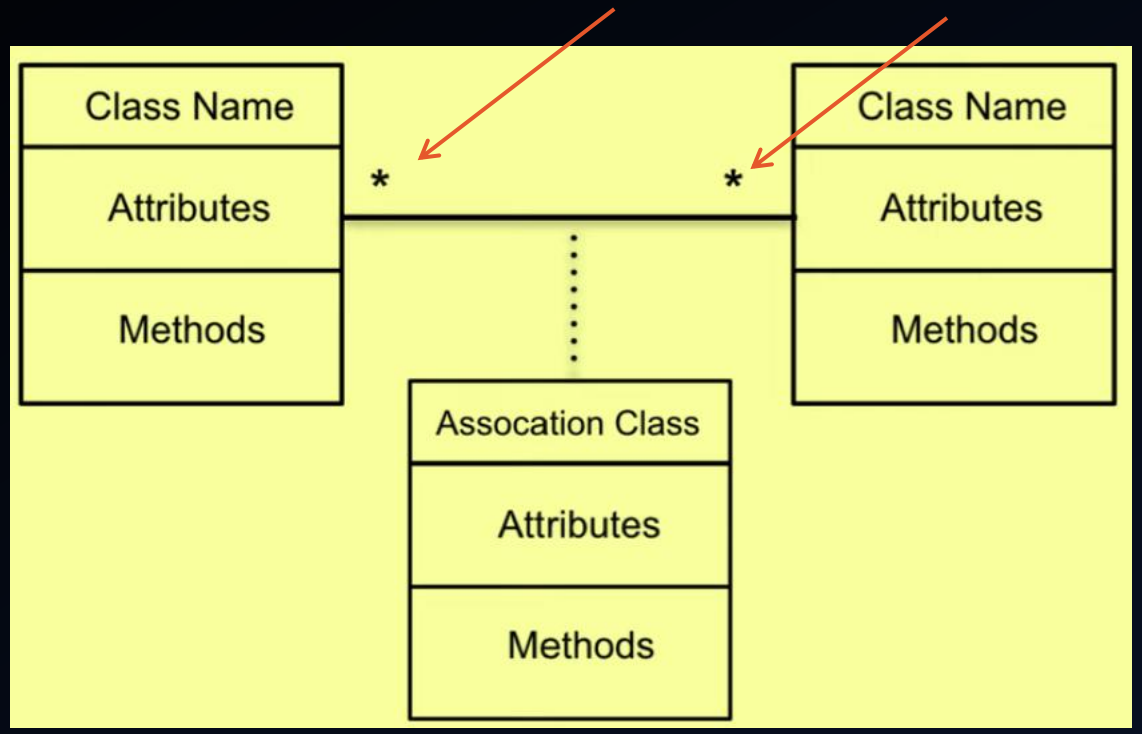
- Manager and Developer both inherit from the Employee class.



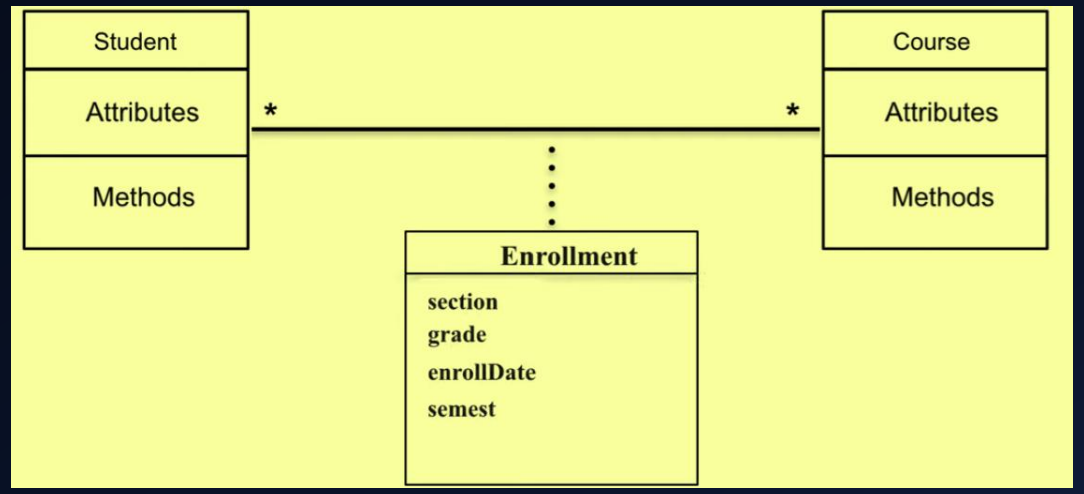
Most of the cases, Parent or superclass is defined as an abstract class << >>

A relationship where a child class inherits features from a parent class.

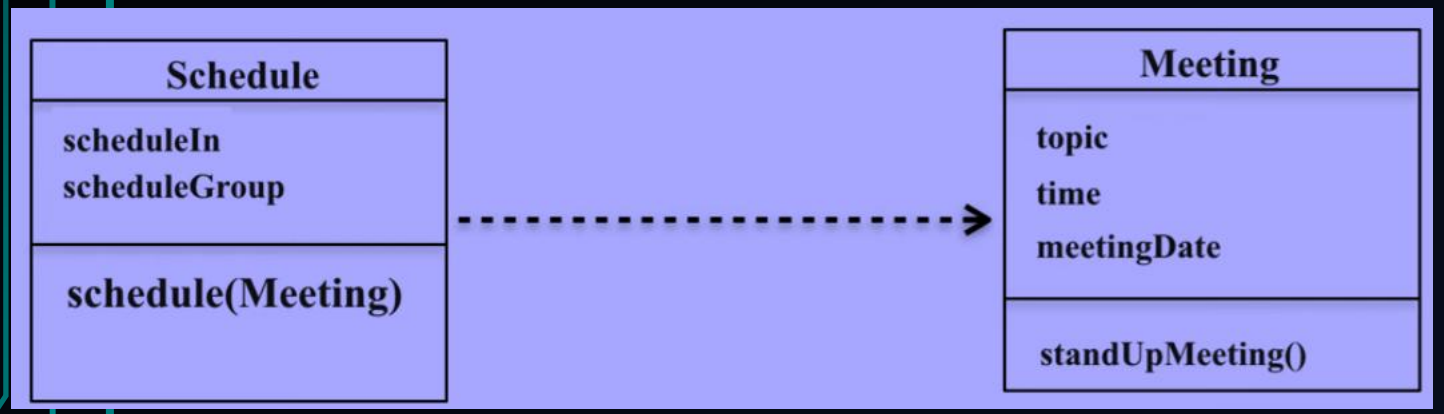
Association Class



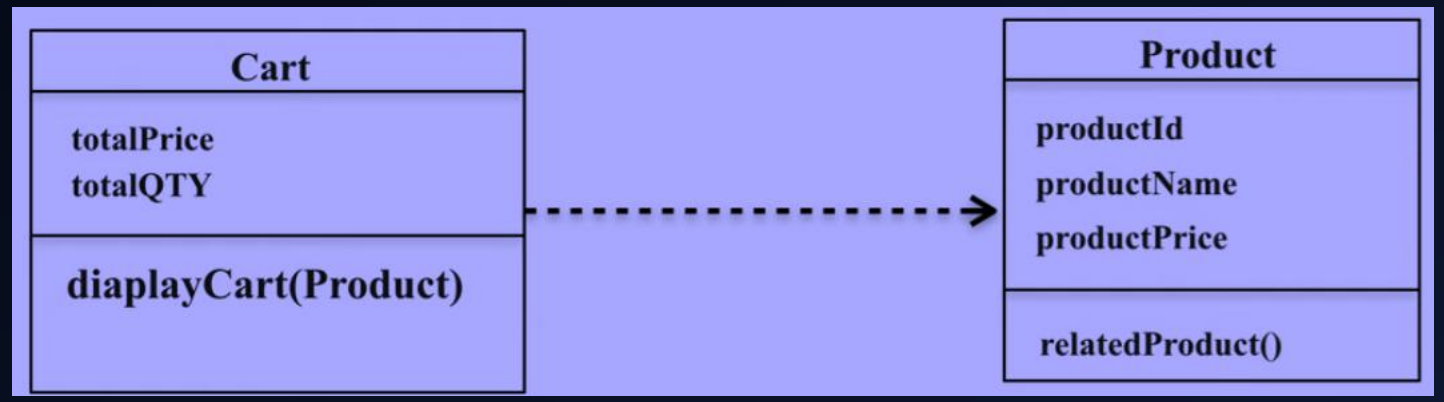
A Student and a Course are linked by an Enrolment class, which stores the "Grade."



Dependency



A weak "uses" relationship. If Class A changes, Class B might need updates. (Dashed arrow).



Task

Implement and design at least **5 classes for 3 systems**, incorporating the key relationships we discussed: **Association, Aggregation, Composition, Inheritance, etc.**

System 1: ATM (Automated Teller Machine) : This system focuses on secure transactions and hardware-software interaction.

System 2: Medical Clinic System: This system focuses on the relationships among staff, patients, and medical records.

System 3: Car Insurance System: relationships between the customer and the company

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Remember to update your SRC Documents to include your topic with a Class diagram.